1. Your job is to synchronize the clocks in a reference frame. You are going to do so by flashing a light at the origin at t = 0 s. To what time should the clock at (x,y,z) = (30 m, 40 m, 0 m) be preset?

2. Bjorn is standing at x = 600 m. Firecracker 1 explodes at the origin, and Firecracker 2 explodes at x = 900 m. The flashes from both explosions reach Bjorn's eye at t = 3 μs. At what time did each explode? Are the two events simultaneous?

3. Bianca is standing at x = 600 m. Firecracker 1, at the origin and firecracker 2, at x = 900 m, explode simultaneously. The flash from firecracker 1 reaches Bianca's eye at t = 3.0 μs. (a) When does the flash from firecracker 2 reach Bianca's eye? (b) What is the actual spacetime coordinates of the two events?

4. You are flying your personal rocketcraft at 0.90c from Star A toward Star B. The distance between the stars, in the stars' reference frame, is 1.0 Ly. Both stars happen to explode simultaneously in your reference frame at the instant you are exactly halfway between them. Do you see the flashes simultaneously? If not which do you see first, and what is the time difference between the two?

5. A radioactive element is sitting in a laboratory when it decays into two fragments. Fragment A heads to the left at 0.8c, while fragment B heads to the right at 0.7 c. What is the velocity of fragment A from fragment B?

6. An alien spaceship is traveling toward the Earth at 0.99c when the Earth shoots a laser at it. Use the velocity addition formula to show that the ship still sees the laser approach it at c.